

Adaptive Body Controller (ABC) — High-Level Design

Body & Comfort Domain — AUTOSAR Classic Platform

Field	Value
Document ID	ABC-HLD-SYN-001
Version	1.0.0
Status	Released
Organization	Synthetic Demonstration Program (fictional)
Security classification	Synthetic sample

This document is a synthetic high-level design generated for an AI engineering-assistant demonstration. It does not describe any real vehicle program.

1 Introduction

1.1 Purpose

This High-Level Design (HLD) describes the software architecture of the Adaptive Body Controller (ABC) for the Body & Comfort domain. It defines software components, their ports and interfaces, communication signals, dependencies, and end-to-end functional flows. Section 3 defines the component architecture; Section 4 specifies interfaces; Section 5 lists the signal dictionary; Section 6 records dependencies; Section 7 describes functional flows.

1.2 Scope

The document covers the application software components of the ABC ECU, the AUTOSAR BSW services they use, and the CAN communication path. Detailed timing analysis, AUTOSAR OS configuration, and ECU resource files are out of scope and are maintained separately.

1.3 Revision History

Version	Date	Author	Changes
1.0.0	2026-08-14	R. Sharma	Initial release of the Adaptive Body Controller HLD.

1.4 References

- [R1] AUTOSAR Classic Platform Release R22-11 documentation.
[R2] ABC System Requirements Specification (synthetic).
[R3] Body & Comfort CAN Communication Matrix (synthetic).

1.5 Terminology

Term	Meaning
SWC	Software Component (AUTOSAR application component)
BSW	Basic Software
S-R interface	Sender-Receiver interface
C-S interface	Client-Server interface
P-port	Provided port (component provides the interface)
R-port	Required port (component requires the interface)

2 System Overview

The ABC ECU implements body and comfort functions. Application SWCs implement feature logic, the RTE connects application ports to communication services, and the BSW stack transmits signals over CAN. The BodyControlSWC (C-08) is the central arbiter: application components submit requests to it, and it releases commands only when vehicle mode (C-09) and vehicle speed (C-10) interlocks permit.

The layered view is: Application layer (C-01..C-10), RTE adapter (C-11), Services layer (C-12, C-13, C-14, C-16, C-17, C-18, C-19), and ECU abstraction (C-15, C-20).

The communication path is: application SWC -> RteAdapter -> Com -> PduR -> CanIf -> CanDriver.

3 Component Architecture

3.1 Component Catalogue

ID	Component	Type	Layer	Description
C-01	DoorControlSWC	SWC	Application	Controls door lock, unlock, and child-lock actuation based on switch inputs and vehicle speed constraints.
C-02	WindowLiftSWC	SWC	Application	Manages one-touch window lift, anti-pinch protection, and window position learning.
C-03	MirrorAdjustSWC	SWC	Application	Controls mirror folding, heating, and position adjustment requests.
C-04	LightControlSWC	SWC	Application	Controls interior ambient lighting and welcome light sequences.
C-05	SeatAdjustSWC	SWC	Application	Handles seat position and lumbar adjustment requests with memory profiles.
C-06	ClimateInterfaceSWC	SWC	Application	Bridges body domain functions to climate blower and recirculation actuators.
C-07	KeylessEntrySWC	SWC	Application	Processes keyless authentication challenges and unlocks authorization.
C-08	BodyControlSWC	SWC	Application	Central body domain arbiter; resolves actuator command conflicts and applies safety interlocks.
C-09	VehicleModeSWC	SWC	Application	Maintains vehicle mode state (park, drive, crash) and distributes mode changes.
C-10	SpeedProviderSWC	SWC	Application	Provides filtered vehicle speed for anti-pinch and child-lock interlocks.
C-11	RteAdapter	RTE	RTE	RTE adapter layer mapping SWC ports onto BSW communication services.
C-12	Com	BSW	Service	AUTOSAR COM module; signal packing, transmission cycles, and timeout monitoring.
C-13	PduR	BSW	Service	PDU router; routes PDUs between COM, CanIf, and transport layers.
C-14	CanIf	BSW	Service	CAN interface; multiplexes PDUs onto CAN controller channels.
C-15	CanDriver	BSW	ECU-Abstraction	CAN driver for the body controller CAN channel with hardware filter configuration.
C-16	CanSm	BSW	Service	CAN state manager; coordinates bus-off recovery and channel start/stop.
C-17	NvM	BSW	Service	NVRAM manager; persists window pinch thresholds and seat memory profiles.
C-18	EcuM	BSW	Service	ECU state manager; coordinates startup, shutdown, and sleep transitions.
C-19	DiagManager	BSW	Service	Unified diagnostic services manager for UDS-based diagnostics.
C-20	IoHwAb	ECU-Abstraction	ECU-Abstraction	I/O hardware abstraction mapping physical pins to logical signals (door switches, window motors, mirror motors).

3.2 Component Details

3.2.1 Component C-01: DoorControlSWC

DoorControlSWC (SWC, Application layer). Controls door lock, unlock, and child-lock actuation based on switch inputs and vehicle speed constraints.

Port	Direction	Interface	Kind
P-001	provides	DoorStatusIF	S-R
P-004	requires	DoorCommandIF	C-S
P-025	requires	VehicleModelIF	S-R
P-028	requires	VehicleSpeedIF	S-R
P-051	requires	IoHwDoorIF	S-R

3.2.2 Component C-02: WindowLiftSWC

WindowLiftSWC (SWC, Application layer). Manages one-touch window lift, anti-pinch protection, and window position learning.

Port	Direction	Interface	Kind
P-005	provides	WindowStatusIF	S-R
P-008	requires	WindowCommandIF	C-S
P-026	requires	VehicleModelIF	S-R
P-029	requires	VehicleSpeedIF	S-R
P-043	requires	NvBlockIF	C-S
P-052	requires	IoHwDoorIF	S-R

3.2.3 Component C-03: MirrorAdjustSWC

MirrorAdjustSWC (SWC, Application layer). Controls mirror folding, heating, and position adjustment requests.

Port	Direction	Interface	Kind
P-009	provides	MirrorStatusIF	S-R
P-012	requires	MirrorCommandIF	C-S
P-054	requires	IoHwMirrorSeatIF	S-R

3.2.4 Component C-04: LightControlSWC

LightControlSWC (SWC, Application layer). Controls interior ambient lighting and welcome light sequences.

Port	Direction	Interface	Kind
P-013	provides	InteriorLightIF	S-R
P-057	requires	IoHwLightIF	S-R

3.2.5 Component C-05: SeatAdjustSWC

SeatAdjustSWC (SWC, Application layer). Handles seat position and lumbar adjustment requests with memory profiles.

Port	Direction	Interface	Kind
P-015	provides	SeatStatusIF	S-R
P-018	requires	SeatCommandIF	C-S
P-044	requires	NvBlockIF	C-S
P-055	requires	IoHwMirrorSeatIF	S-R

3.2.6 Component C-06: ClimateInterfaceSWC

ClimateInterfaceSWC (SWC, Application layer). Bridges body domain functions to climate blower and recirculation actuators.

Port	Direction	Interface	Kind
P-019	provides	ClimateRequestIF	C-S

3.2.7 Component C-07: KeylessEntrySWC

KeylessEntrySWC (SWC, Application layer). Processes keyless authentication challenges and unlocks authorization.

Port	Direction	Interface	Kind
P-021	provides	KeyAuthIF	S-R

3.2.8 Component C-08: BodyControlSWC

BodyControlSWC (SWC, Application layer). Central body domain arbiter; resolves actuator command conflicts and applies safety interlocks.

Port	Direction	Interface	Kind
P-002	requires	DoorStatusIF	S-R
P-003	provides	DoorCommandIF	C-S
P-006	requires	WindowStatusIF	S-R
P-007	provides	WindowCommandIF	C-S
P-010	requires	MirrorStatusIF	S-R
P-011	provides	MirrorCommandIF	C-S
P-014	requires	InteriorLightIF	S-R
P-016	requires	SeatStatusIF	S-R
P-017	provides	SeatCommandIF	C-S
P-020	requires	ClimateRequestIF	C-S
P-022	requires	KeyAuthIF	S-R
P-024	requires	VehicleModelIF	S-R

3.2.9 Component C-09: VehicleModeSWC

VehicleModeSWC (SWC, Application layer). Maintains vehicle mode state (park, drive, crash) and distributes mode changes.

Port	Direction	Interface	Kind
P-023	provides	VehicleModelIF	S-R

3.2.10 Component C-10: SpeedProviderSWC

SpeedProviderSWC (SWC, Application layer). Provides filtered vehicle speed for anti-pinch and child-lock interlocks.

Port	Direction	Interface	Kind
P-027	provides	VehicleSpeedIF	S-R

3.2.11 Component C-11: RteAdapter

RteAdapter (RTE, RTE layer). RTE adapter layer mapping SWC ports onto BSW communication services.

Port	Direction	Interface	Kind
P-030	provides	RteComIF	S-R
P-033	requires	RtePduIF	C-S
P-049	requires	DiagRequestIF	C-S

3.2.12 Component C-12: Com

Com (BSW, Service layer). AUTOSAR COM module; signal packing, transmission cycles, and timeout monitoring.

Port	Direction	Interface	Kind
P-031	requires	RteComIF	S-R
P-032	provides	RtePduIF	C-S
P-034	provides	ComPduRIF	C-S
P-047	requires	EcuModelIF	S-R

3.2.13 Component C-13: PduR

PduR (BSW, Service layer). PDU router; routes PDUs between COM, CanIf, and transport layers.

Port	Direction	Interface	Kind
P-035	requires	ComPduRIF	C-S
P-036	provides	PduCanIfIF	C-S

3.2.14 Component C-14: CanIf

CanIf (BSW, Service layer). CAN interface; multiplexes PDUs onto CAN controller channels.

Port	Direction	Interface	Kind
P-037	requires	PduCanIfIF	C-S
P-038	provides	CanIfDriverIF	C-S

3.2.15 Component C-15: CanDriver

CanDriver (BSW, ECU-Abstraction layer). CAN driver for the body controller CAN channel with hardware filter configuration.

Port	Direction	Interface	Kind
P-039	requires	CanIfDriverIF	C-S
P-041	requires	CanSmControlIF	C-S

3.2.16 Component C-16: CanSm

CanSm (BSW, Service layer). CAN state manager; coordinates bus-off recovery and channel start/stop.

Port	Direction	Interface	Kind
P-040	provides	CanSmControlIF	C-S
P-046	requires	EcuModelIF	S-R

3.2.17 Component C-17: NvM

NvM (BSW, Service layer). NVRAM manager; persists window pinch thresholds and seat memory profiles.

Port	Direction	Interface	Kind
P-042	provides	NvBlockIF	C-S

3.2.18 Component C-18: EcuM

EcuM (BSW, Service layer). ECU state manager; coordinates startup, shutdown, and sleep transitions.

Port	Direction	Interface	Kind
P-045	provides	EcuModelIF	S-R

3.2.19 Component C-19: DiagManager

DiagManager (BSW, Service layer). Unified diagnostic services manager for UDS-based diagnostics.

Port	Direction	Interface	Kind
P-048	provides	DiagRequestIF	C-S

3.2.20 Component C-20: IoHwAb

IoHwAb (ECU-Abstraction, ECU-Abstraction layer). I/O hardware abstraction mapping physical pins to logical signals (door switches, window motors, mirror motors).

Port	Direction	Interface	Kind
P-050	provides	IoHwDoorIF	S-R
P-053	provides	IoHwMirrorSeatIF	S-R
P-056	provides	IoHwLightIF	S-R

4 Interface Specifications

Each interface below lists its provider (P-port owner) and its consumers (R-port owners). Signal definitions for each interface are given in Section 5.

4.1 DoorStatusIF (IF-01)

DoorStatusIF is a S-R interface. Provider: C-01 (DoorControlSWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
DoorStatus	uint8	-
ChildLockActive	boolean	-

4.2 DoorCommandIF (IF-02)

DoorCommandIF is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-01 (DoorControlSWC).

Signal	Datatype	Unit
DoorLockRequest	uint8	-

4.3 WindowStatusIF (IF-03)

WindowStatusIF is a S-R interface. Provider: C-02 (WindowLiftSWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
WindowPosition	uint8	%
AntiPinchTriggered	boolean	-

4.4 WindowCommandIF (IF-04)

WindowCommandIF is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-02 (WindowLiftSWC).

Signal	Datatype	Unit
WindowLiftCommand	sint8	-

4.5 MirrorStatusIF (IF-05)

MirrorStatusIF is a S-R interface. Provider: C-03 (MirrorAdjustSWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
MirrorFoldState	boolean	-
MirrorPosition	uint8	deg

4.6 MirrorCommandIF (IF-06)

MirrorCommandIF is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-03 (MirrorAdjustSWC).

Signal	Datatype	Unit
MirrorAdjustCommand	uint8	-

4.7 InteriorLightIF (IF-07)

InteriorLightIF is a S-R interface. Provider: C-04 (LightControlSWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
InteriorLightLevel	uint8	%

4.8 SeatStatusIF (IF-08)

SeatStatusIF is a S-R interface. Provider: C-05 (SeatAdjustSWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
SeatPosition	uint16	mm
SeatMemorySlot	uint8	-

4.9 SeatCommandIF (IF-09)

SeatCommandIF is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-05 (SeatAdjustSWC).

Signal	Datatype	Unit
SeatAdjustCommand	sint8	-

4.10 ClimateRequestIF (IF-10)

ClimateRequestIF is a C-S interface. Provider: C-06 (ClimateInterfaceSWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
BlowerLevelRequest	uint8	%

4.11 KeyAuthIF (IF-11)

KeyAuthIF is a S-R interface. Provider: C-07 (KeylessEntrySWC). Consumers: C-08 (BodyControlSWC).

Signal	Datatype	Unit
KeyAuthStatus	uint8	-

4.12 VehicleModelIF (IF-12)

VehicleModelIF is a S-R interface. Provider: C-09 (VehicleModeSWC). Consumers: C-08 (BodyControlSWC), C-01 (DoorControlSWC), C-02 (WindowLiftSWC).

Signal	Datatype	Unit
VehicleMode	uint8	-
CrashSignal	boolean	-

4.13 VehicleSpeedIF (IF-13)

VehicleSpeedIF is a S-R interface. Provider: C-10 (SpeedProviderSWC). Consumers: C-01 (DoorControlSWC), C-02 (WindowLiftSWC).

Signal	Datatype	Unit
VehicleSpeed	uint16	km/h
SpeedValidity	boolean	-

4.14 RteComIF (IF-14)

RteComIF is a S-R interface. Provider: C-11 (RteAdapter). Consumers: C-12 (Com).

Signal	Datatype	Unit
ComSignalBatch1	uint8[8]	-

4.15 RtePduIF (IF-15)

RtePduIF is a C-S interface. Provider: C-12 (Com). Consumers: C-11 (RteAdapter).

Signal	Datatype	Unit
ComSignalBatch2	uint8[8]	-

4.16 ComPduRIF (IF-16)

ComPduRIF is a C-S interface. Provider: C-12 (Com). Consumers: C-13 (PduR).

Signal	Datatype	Unit
PduPayload	uint8[64]	-

4.17 PduCanIfIF (IF-17)

PduCanIfIF is a C-S interface. Provider: C-13 (PduR). Consumers: C-14 (CanIf).

Signal	Datatype	Unit
CanFrame	uint8[8]	-

4.18 CanIfDriverIF (IF-18)

CanIfDriverIF is a C-S interface. Provider: C-14 (CanIf). Consumers: C-15 (CanDriver).

Signal	Datatype	Unit
CanFrameRaw	uint8[8]	-

4.19 CanSmControlIF (IF-19)

CanSmControlIF is a C-S interface. Provider: C-16 (CanSm). Consumers: C-15 (CanDriver).

Signal	Datatype	Unit
CanChannelState	uint8	-

4.20 NvBlockIF (IF-20)

NvBlockIF is a C-S interface. Provider: C-17 (NvM). Consumers: C-02 (WindowLiftSWC), C-05 (SeatAdjustSWC).

Signal	Datatype	Unit
NvBlockStatus	uint8	-

4.21 EcuModelIF (IF-21)

EcuModelIF is a S-R interface. Provider: C-18 (EcuM). Consumers: C-16 (CanSm), C-12 (Com).

Signal	Datatype	Unit
EcuModeState	uint8	-

4.22 DiagRequestIF (IF-22)

DiagRequestIF is a C-S interface. Provider: C-19 (DiagManager). Consumers: C-11 (RteAdapter).

Signal	Datatype	Unit
DiagRequestPayload	uint8[255]	-

4.23 IoHwDoorIF (IF-23)

IoHwDoorIF is a S-R interface. Provider: C-20 (IoHwAb). Consumers: C-01 (DoorControlSWC), C-02 (WindowLiftSWC).

Signal	Datatype	Unit
DoorSwitchRaw	uint8	-
WindowMotorCurrent	uint16	mA

4.24 IoHwMirrorSeatIF (IF-24)

IoHwMirrorSeatIF is a S-R interface. Provider: C-20 (IoHwAb). Consumers: C-03 (MirrorAdjustSWC), C-05 (SeatAdjustSWC).

Signal	Datatype	Unit
MirrorMotorFeedback	uint16	mA
SeatMotorFeedback	uint16	mA

4.25 IoHwLightIF (IF-25)

IoHwLightIF is a S-R interface. Provider: C-20 (IoHwAb). Consumers: C-04 (LightControlSWC).

Signal	Datatype	Unit
LightSwitchRaw	uint8	-
AmbientLightFeedback	uint8	%

5 Signal Dictionary

The signal dictionary lists all signals carried by the interfaces of Section 4, including datatype and physical unit. The VehicleSpeed signal (SG-018) is filtered by the SpeedProviderSWC before distribution.

Signal ID	Signal	Datatype	Unit	Interface
SG-001	DoorStatus	uint8	-	DoorStatusIF
SG-002	ChildLockActive	boolean	-	DoorStatusIF
SG-003	DoorLockRequest	uint8	-	DoorCommandIF
SG-004	WindowPosition	uint8	%	WindowStatusIF
SG-005	AntiPinchTriggered	boolean	-	WindowStatusIF
SG-006	WindowLiftCommand	sint8	-	WindowCommandIF
SG-007	MirrorFoldState	boolean	-	MirrorStatusIF
SG-008	MirrorPosition	uint8	deg	MirrorStatusIF
SG-009	MirrorAdjustCommand	uint8	-	MirrorCommandIF
SG-010	InteriorLightLevel	uint8	%	InteriorLightIF
SG-011	SeatPosition	uint16	mm	SeatStatusIF
SG-012	SeatMemorySlot	uint8	-	SeatStatusIF
SG-013	SeatAdjustCommand	sint8	-	SeatCommandIF
SG-014	BlowerLevelRequest	uint8	%	ClimateRequestIF
SG-015	KeyAuthStatus	uint8	-	KeyAuthIF
SG-016	VehicleMode	uint8	-	VehicleModelIF
SG-017	CrashSignal	boolean	-	VehicleModelIF
SG-018	VehicleSpeed	uint16	km/h	VehicleSpeedIF
SG-019	SpeedValidity	boolean	-	VehicleSpeedIF
SG-020	ComSignalBatch1	uint8[8]	-	RteComIF
SG-021	ComSignalBatch2	uint8[8]	-	RtePduIF
SG-022	PduPayload	uint8[64]	-	ComPduRIF
SG-023	CanFrame	uint8[8]	-	PduCanIfIF
SG-024	CanFrameRaw	uint8[8]	-	CanIfDriverIF
SG-025	CanChannelState	uint8	-	CanSmControlIF
SG-026	NvBlockStatus	uint8	-	NvBlockIF
SG-027	EcuModeState	uint8	-	EcuModelIF
SG-028	DiagRequestPayload	uint8[255]	-	DiagRequestIF
SG-029	DoorSwitchRaw	uint8	-	IoHwDoorIF
SG-030	WindowMotorCurrent	uint16	mA	IoHwDoorIF
SG-031	MirrorMotorFeedback	uint16	mA	IoHwMirrorSeatIF
SG-032	SeatMotorFeedback	uint16	mA	IoHwMirrorSeatIF
SG-033	LightSwitchRaw	uint8	-	IoHwLightIF
SG-034	AmbientLightFeedback	uint8	%	IoHwLightIF

6 Dependencies

6.1 Dependency Overview

Dependencies record which components require services or commands from other components. Every dependency in this section is realized by an interface defined in Section 4.

Dep. ID	Source	Target	Type
DEP-01	C-01 (DoorControlSWC)	C-08 (BodyControlSWC)	requires
DEP-02	C-02 (WindowLiftSWC)	C-08 (BodyControlSWC)	requires
DEP-03	C-03 (MirrorAdjustSWC)	C-08 (BodyControlSWC)	requires
DEP-04	C-05 (SeatAdjustSWC)	C-08 (BodyControlSWC)	requires
DEP-05	C-06 (ClimateInterfaceSWC)	C-08 (BodyControlSWC)	requires
DEP-06	C-08 (BodyControlSWC)	C-09 (VehicleModeSWC)	requires
DEP-07	C-08 (BodyControlSWC)	C-10 (SpeedProviderSWC)	requires
DEP-08	C-02 (WindowLiftSWC)	C-10 (SpeedProviderSWC)	requires
DEP-09	C-01 (DoorControlSWC)	C-07 (KeylessEntrySWC)	requires
DEP-10	C-11 (RteAdapter)	C-12 (Com)	depends_on
DEP-11	C-12 (Com)	C-13 (PduR)	depends_on
DEP-12	C-13 (PduR)	C-14 (CanIf)	depends_on
DEP-13	C-14 (CanIf)	C-15 (CanDriver)	depends_on
DEP-14	C-16 (CanSm)	C-15 (CanDriver)	requires
DEP-15	C-02 (WindowLiftSWC)	C-17 (NvM)	requires
DEP-16	C-05 (SeatAdjustSWC)	C-17 (NvM)	requires
DEP-17	C-18 (EcuM)	C-16 (CanSm)	requires
DEP-18	C-18 (EcuM)	C-12 (Com)	requires
DEP-19	C-19 (DiagManager)	C-11 (RteAdapter)	requires
DEP-20	C-01 (DoorControlSWC)	C-20 (IoHwAb)	requires
DEP-21	C-02 (WindowLiftSWC)	C-20 (IoHwAb)	requires
DEP-22	C-03 (MirrorAdjustSWC)	C-20 (IoHwAb)	requires
DEP-23	C-04 (LightControlSWC)	C-20 (IoHwAb)	requires
DEP-24	C-05 (SeatAdjustSWC)	C-20 (IoHwAb)	requires

6.2 Dependency Details

6.2.1 DEP-01

The DoorControlSWC requires door lock and unlock commands from the BodyControlSWC.

6.2.2 DEP-02

The WindowLiftSWC requires window lift commands from the BodyControlSWC.

6.2.3 DEP-03

The MirrorAdjustSWC requires mirror adjustment commands from the BodyControlSWC.

6.2.4 DEP-04

The SeatAdjustSWC requires seat adjustment commands from the BodyControlSWC.

6.2.5 DEP-05

The ClimateInterfaceSWC requires blower control authorization from the BodyControlSWC.

6.2.6 DEP-06

The BodyControlSWC requires vehicle mode information from the VehicleModeSWC before releasing actuator commands.

6.2.7 DEP-07

The BodyControlSWC requires filtered vehicle speed from the SpeedProviderSWC for anti-pinch interlocks.

6.2.8 DEP-08

The WindowLiftSWC requires vehicle speed to qualify one-touch window operation.

6.2.9 DEP-09

The DoorControlSWC requires keyless authorization status from the KeylessEntrySWC.

6.2.10 DEP-10

The RteAdapter depends on the COM module for signal transmission.

6.2.11 DEP-11

The COM module depends on the PduR for PDU routing.

6.2.12 DEP-12

The PduR depends on the CanIf for CAN transmission.

6.2.13 DEP-13

The CanIf depends on the CanDriver for hardware access.

6.2.14 DEP-14

The CanSm requires bus state control of the CanDriver.

6.2.15 DEP-15

The WindowLiftSWC requires NVRAM persistence of anti-pinch thresholds from the NvM.

6.2.16 DEP-16

The SeatAdjustSWC requires NVRAM persistence of memory profiles from the NvM.

6.2.17 DEP-17

The EcuM requires communication channel control from the CanSm during shutdown.

6.2.18 DEP-18

The EcuM requires COM transmission suspension during shutdown.

6.2.19 DEP-19

The DiagManager requires the RteAdapter for diagnostic read access to application data.

6.2.20 DEP-20

The DoorControlSWC requires raw door switch inputs from the IoHwAb.

6.2.21 DEP-21

The WindowLiftSWC requires window motor current feedback from the IoHwAb.

6.2.22 DEP-22

The MirrorAdjustSWC requires mirror motor feedback from the IoHwAb.

6.2.23 DEP-23

The LightControlSWC requires ambient light feedback from the IoHwAb.

6.2.24 DEP-24

The SeatAdjustSWC requires seat motor feedback from the IoHwAb.

7 Functional Flows

Functional flows describe end-to-end behaviour across components. Flow steps reference the components of Section 3.

7.1 Door Lock Actuation (FL-1)

Trigger: Driver operates door lock switch

Flow: C-20 (IoHwAb) → C-01 (DoorControlSWC) → C-08 (BodyControlSWC) → C-11 (RteAdapter) → C-12 (Com) → C-13 (PduR) → C-14 (CanIf) → C-15 (CanDriver)

Raw switch input is debounced in IoHwAb, evaluated by DoorControlSWC, arbitrated by BodyControlSWC, and the resulting command is transmitted over CAN.

7.2 One-Touch Window Lift with Anti-Pinch (FL-2)

Trigger: Driver triggers one-touch window lift

Flow: C-20 (IoHwAb) → C-02 (WindowLiftSWC) → C-10 (SpeedProviderSWC) → C-08 (BodyControlSWC) → C-11 (RteAdapter) → C-12 (Com) → C-13 (PduR) → C-14 (CanIf) → C-15 (CanDriver)

Motor current from IoHwAb is monitored by WindowLiftSWC; vehicle speed from SpeedProviderSWC qualifies the command; BodyControlSWC arbitrates and forwards the request over CAN.

7.3 Keyless Unlock (FL-3)

Trigger: Authenticated key approach detected

Flow: C-07 (KeylessEntrySWC) → C-08 (BodyControlSWC) → C-01 (DoorControlSWC) → C-20 (IoHwAb)

KeylessEntrySWC publishes the authentication result; BodyControlSWC authorizes the unlock; DoorControlSWC actuates via IoHwAb.

7.4 Seat Memory Recall (FL-4)

Trigger: Driver selects memory slot

Flow: C-17 (NvM) → C-05 (SeatAdjustSWC) → C-08 (BodyControlSWC) → C-20 (IoHwAb)

NvM provides the persisted profile, SeatAdjustSWC computes motor targets, BodyControlSWC arbitrates, and IoHwAb drives the seat motors.

7.5 Bus-Off Recovery (FL-5)

Trigger: CAN controller enters bus-off state

Flow: C-15 (CanDriver) → C-16 (CanSm) → C-18 (EcuM)

CanDriver reports the fault, CanSm executes the recovery sequence, and EcuM is informed for mode bookkeeping.

8 Assumptions and Open Points

8.1 Assumptions

- A1. The CAN bus operates at 500 kbit/s.
- A2. The NvM stores up to three seat memory profiles per user.
- A3. Anti-pinch protection is mandatory for all windows when vehicle speed is below 5 km/h.

8.2 Open Points

- OP-1: Diagnostic read access for application data is handled via the DiagManager, which requires the RteAdapter for data access (see DEP-19).
- OP-2: Welcome-light sequencing depends on vehicle mode distribution (IF-12).